



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECP2523 – DATABASE

SEM 1 - 2024/2025

INITIAL SYSTEM PROPOSAL
SYSTEM KADA ESERVE

Group Name: TECH AWAY

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**Stakeholder: Lembaga Kemajuan Pertanian Kemubu
(KADA)**

Representative Name	Date Interviewed
Mr. Ahmad Rohailan Bin Hani	12/5/2024

PROPOSED SYSTEM BACKGROUND

1. Introduction

Efficient document management system (DMS) is essential for every organization since it significantly reduce the time required for document handling and optimizing organizational workflows. However, Lembaga Kemajuan Pertanian Kemubu (KADA) still rely on traditional physical document management systems which makes it difficult for them to keep up with current needs. KADA currently aims to improve service delivery and optimize internal processes, there is a pressing need to transition from traditional methods to digital solutions that offer greater efficiency, security, and accessibility.

Mr. Ahmad Rohailan Bin Hani, KADA's representative, emphasized that the agency's existing reliance on paper-based document management leads to sluggish processing, ineffective workflows, and a higher chance of document loss. Additionally, the absence of strong reporting capabilities restricts the ability to make decisions based on data, which makes it more difficult for them to adjust to the quick changes in information management. Therefore, KADA needs us to create an effective DMS system that able to solve these restrictions.

In this proposal, we have introduced our system named KADA eServe that is designed to improve the document management efficiency and accuracy of document and workflow management. The system will digitalize the form application for loan and member register, accelerate document processing and minimizing risk of document loss. Otherwise, reporting feature will be introduced in the system for more convenience to refer back previous records. In addition, local database will be used in the system for storing and retrieving data immediately while using the system.

2. The Existing System

2.1 AS-IS System

Currently, KADA's existing system (as-is system) facing lots of difficulty in managing member registration, loan applications and member financial statement. Efficiency, accuracy and accessibility are all impacted by the manual method using by the company. Below are the key aspects of the existing system and the problem faced by the client while using the current system.

For member registration process, applicants have to complete out hardcopy registration forms with their information such as name, identity card number, salary, and contribution amount. Following this procedure, applicants must get managers' signatures and cross-check the information for completeness. Otherwise, KADA uses a paper-based procedure for loan applications. Under this procedure, candidates must fill out paper application forms, which KADA staff members then manually process.

Moreover, KADA currently generates and distributes member financial statements using a manual approach. This involves manually gathering financial statements from multiple sources and assembling them into statements for each member. Important information such as shares held, loans carried out, and savings must

be included in the member statement. When candidates submit their information, KADA staff members check it for accuracy.

The current website for KADA lacks the functionality found in the manual system, including those that are typically managed manually. These missing features include member registration, loan application, member financial system and automated generation of reports. Other than that, using manually applications may require a huge office space for data storing. KADA's company does not using database for storing forms information and member financial statement. As a result, KADA's staff and members face significant inefficiencies in managing day-to-day operations and accessing essential services.

2.2 Problem Statement

KADA currently relies heavily on manual, paper-based processes for member registration, loan applications, and data management. This traditional approach has led to several challenges that hinder operational efficiency and the organization's ability to adapt to modern demands.

1. **Inefficiency in Member Registration and Data Entry**

The manual registration process requires users to fill out paper forms, which requiring staff must then process and input into the system manually. This process is time-consuming and prone to errors, such as incorrect data input or duplication of records. These inefficiencies result in inconsistent data integrity and delays in updating the database, which affects overall system responsiveness and member onboarding timelines.

2. **Limited Data Storage Capacity**

KADA's current digital storage capacity for member registration is limited to on a few hundred members, posing a significant obstacle to the organization's expansion. The inability to handle a larger volume of data restricts KADA's ability to recruit and manage more members. This limitation also affects the organization's capacity to maintain comprehensive records, generate detailed reports, and conduct effective data analysis, all of which are vital for strategic planning and operational efficiency.

3. **Security**

The reliance on manual data entry and paper forms in KADA's current system presents significant security risks. Without proper encryption and access controls, the database is exposed to vulnerabilities, and manual entry increases the errors such as incorrect or incomplete records. These gaps compromise data confidentiality and operational efficiency.

4. **Lack of Resources to Support Decision Making**

The current database system lacks the features necessary to support advanced data analytics and reporting. Data is fragmented and stored in an unstructured format, hindering the ability to generate insightful reports and monitor operational trends. This limitation affects KADA's capacity to leverage historical data for strategic decision-making and hampers its ability to perform predictive analysis for future planning.

3. The Proposed System

The system will be developed and named KADA eServe, a digital solution designed to streamline the operational process of KADA while enhancing user experience. This system has to meet the organization's needs and provides efficient management of user registration, loan applications and administrative.

To address the challenges posed by KADA's reliance on manual processes and limited digital systems, we propose the implementation of a robust, scalable and secure database-driven solution. This proposal outlines how the adoption of a database system can resolve the identified inefficiencies, improve operational efficiency and support strategic growth.

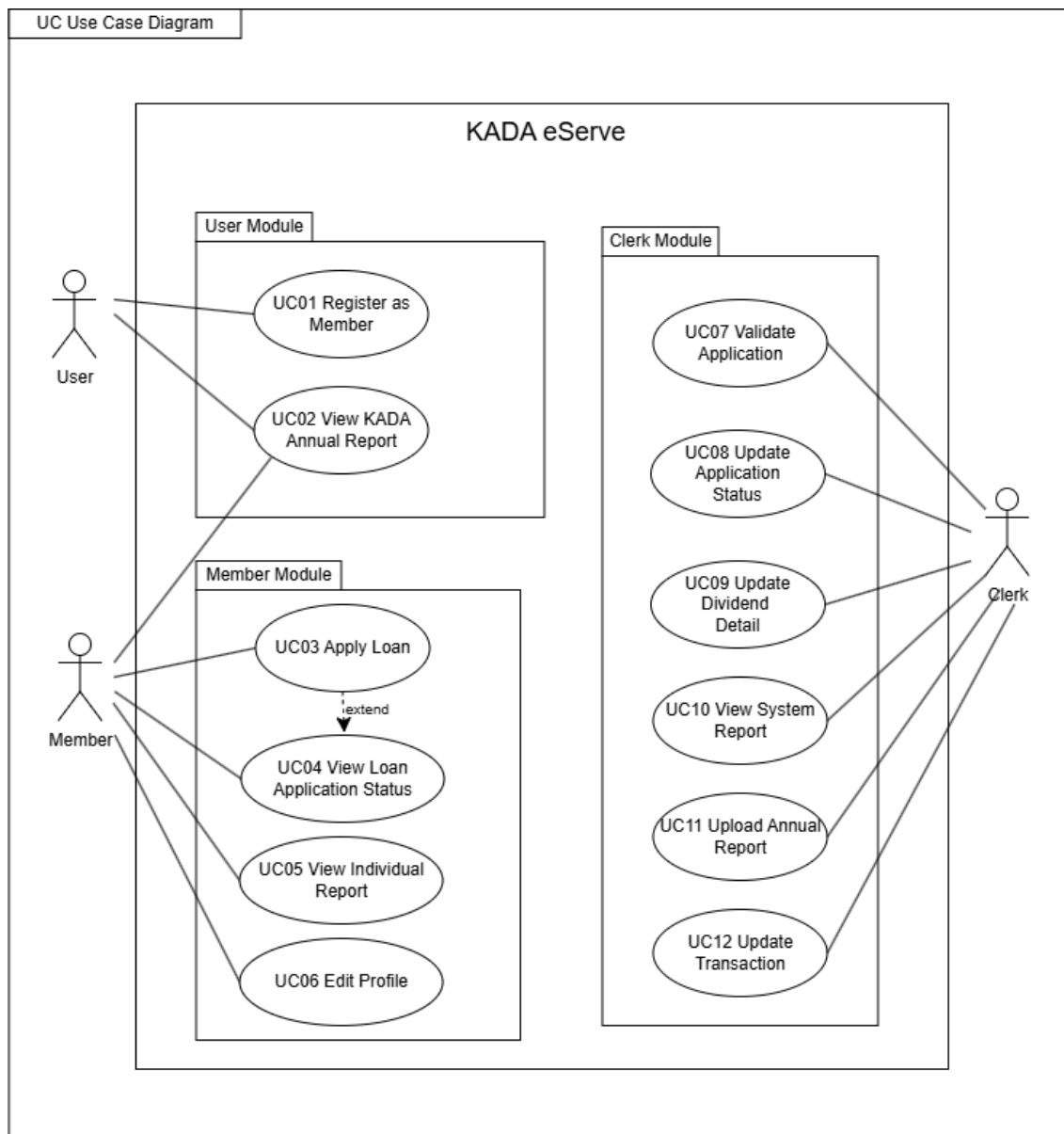
For the first problem that the company faced is addressing inefficiency in member registration and data entry. The inefficiency in member registration can be resolved by implementing an online registration system integrated with a centralized database. Members can complete digital registration forms which are validated and stored directly in the database without the need for manual data entry. Validation rules will also ensure that all inputs are consistent and error-free, reducing the chances of incorrect or duplicate records. By automating the data entry process, this solution streamlines member onboarding, making member data immediately available for other system functionalities.

Second, to overcome the limitations of KADA's current storage capacity, we have implemented a scalable relational database management system (RDBMS) such as MySQL is proposed. This will enable the system to effectively handle thousands of member records and support future expansion. Storage optimisation can be done by normalising data structures which ensuring consistency and cuts down on redundancy. Additionally, a data archiving strategy will ensure that inactive or historical records are stored securely without affecting system performance.

Next, the security risks associated with manual processes can be mitigated by integrating robust security mechanisms into the database system. AES and SSL/TLS encryption protocols will be used to protect sensitive data while it is being transmitted and while it is at rest. To improve confidentiality, role-based access restrictions will make sure that only individuals with permission can view or alter particular data. By taking these steps, the system's security will be greatly improved and member data will be shielded from breaches.

Lastly, a structured relational database will enable efficient querying and analysis of data. Real-time reporting will provide insights into member activities, loan statuses, and organizational performance, empowering administrators to monitor trends effectively.

In conclusion, by implementing these database solutions, KADA can transition to a modern and efficient system that overcomes existing challenges, enhances security, supports scalability and provides actionable insights for informed decision-making.



4. Glossary

- MySQL (Structured Query Language)

MySQL is an open-source relational database management system (RDBMS) that used to store and manage data. In this project, our group will use MySQL to store locally and managing the data.

- HTML (Hyper Text Markup Language)

HTML is a standard markup language for web pages. With HTML you can create your own website.

- Tailwind CSS

Tailwind CSS is a utility-first CSS framework for rapidly building modern websites without ever leaving your HTML.

- PHP (Hypertext Preprocessor)

PHP is a server-side scripting language designed specifically for web development that is free to download and use.

5. Bibliography

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